

Avi Seth

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Roles I am looking for: **Data Scientist, ML Engineer, Python Developer, Data Analyst, Solutions Engineer**

EDUCATION

- **Virginia Tech** Blacksburg, VA
Master of Science in Computer Science (GPA: 4.00/4.00) Jan 2021 – May 2023
 - Thesis: **Data Sharing and Retrieval of Similar Manufacturing Processes**
(Full text: <https://vtechworks.lib.vt.edu/items/5594874e-03b8-4598-8ea5-a0269059a807>)
 - Courses: Advanced ML, Trustworthy ML, Deep Learning, Human-AI Interaction, Project Management
- **NMIMS, Mukesh Patel School of Technology Management & Engineering** Mumbai, India
Bachelor of Technology in Computer Science and Engineering Jul 2016 – May 2020
 - Bachelor's Project: **Efficient Feature Detection from low-res camera sources for Object Identification**
(Full text: <https://shorturl.at/IUyVX>)
 - Courses: Operating Systems, Data Structures, AI, Robotics, Statistics

PROFESSIONAL EXPERIENCE

- **IntraEdge** New York City, NY
Machine Learning Software Engineer Apr 2024 – Present
 - Spearheaded ML/AI projects using AWS SageMaker, Bedrock, and CDK for scalable machine learning pipelines.
 - Applied Gen AI, LLMs, and NLP to address data challenges in recruiting, healthcare, finance, and education.
 - Collaborated with cross-functional teams, including software engineers, data engineers, and domain experts, to ensure seamless integration of ML systems into production environments.
- **The PLAN Lab, Virginia Tech** Blacksburg, VA
Research Assistant (Full Time) May 2023 – April 2024
 - Developed **PriED**, a privacy-preserving data-sharing framework with VAE-LSTM using PyTorch and TensorFlow to optimise model performance for industrial ML tasks.
 - Built a task-specific search engine with bilinear attention and reinforcement learning for data selection.
 - Deployed scalable ML solutions on AWS EC2 using Docker and Kubernetes for distributed experiments.
 - Applied federated learning and differential privacy to ensure data privacy in multi-party collaborations.
- **Gastrograph AI - Analytical Flavor Systems** New York City, NY
Data Scientist Master's Intern May 2022 – Aug 2022
 - Designed and deployed a **food preference generalization model**, improving demographic accuracy from 43% to 84% using ensemble methods (XGBoost and Neural Networks).
 - Applied PyTorch for ICA feature extraction and Bayesian optimization for hyperparameter tuning
 - Validated ensemble methods, enhancing prediction robustness across diverse population datasets.
- **Advanced Research Computing, Virginia Tech** Blacksburg, VA
Graduate Research Assistant - Python Developer Aug 2021 – Jan 2022
 - Developed **OpenOnDemand** apps, improving ML researcher access to HPC resources and boosting user engagement by 30%.
 - Engineered web apps for RStudio, MATLAB, and CAD tools, ensuring low-latency distributed system interactions.
 - Partnered with stakeholders to customize HPC workflows using Python and Bash automation to streamline complex computational tasks.
- **Data Science for the Public Good (DSPG), Virginia Tech** Blacksburg, VA
Graduate Team Leader (Technical Product Management and Data Science) May 2021 – Aug 2021
 - Supervised 15 undergrad interns, collaborated with three graduate leaders, delivered public policy data solutions.
 - Led Hampton Roads data analysis, identifying issues faced by African American communities using R and Tableau.
 - Built a Shiny dashboard for NDVI data, uncovering trends in vegetation indices to inform quality of life policies.
 - Applied statistical methods and visualization techniques to derive actionable insights for policymakers.

RESEARCH EXPERIENCE

- **AI India Innovation Center** Pune, India
Computer Vision Intern Apr 2020 – Jun 2020
 - Compared CV models (TensorFlow, Keras) for computational efficiency on CCTV systems with limited hardware.
 - Tuned hyperparameters and implemented a mask detection and social distancing system.
 - Developed Python-based GUI to integrate live video feeds and real-time stats for policy enforcement.

- **Indian Institute of Technology - Bombay**

Research Intern

Mumbai, India

Aug 2019 – Jan 2020

- Constructed a Bayesian FlowNet-based model using CNNs and LSTMs to enhance video across time intervals.
- Improved drone and CCTV footage by modeling frame transitions for seamless output.
- Conducted experiments validating frame-based predictions for real-world scenarios.

- **Arqaam GmbH**

Data Scientist Intern

Hamburg, Germany

Apr 2019 – Jul 2019

- Analyzed 11500 survey responses from war-torn Middle Eastern zones, converting data to numerical insights.
- Built a statistical model to assess human development and livelihood indices for actionable policy decisions.
- Designed R-based scripts to automate data cleaning, feature engineering, and trend analysis.

CONFERENCE AND PROJECT PAPERS

- **Task-Driven Privacy-Preserving Data-Sharing Framework for the Industrial Internet** (Active Learning, ML, VAE-LSTM)

Presented a framework that selectively combines shared and local data to enhance supervised learning.

(Full text: <https://ieeexplore.ieee.org/document/10020861>)

- **Dashboards OnDemand**

Presented at ISAIC 2021 (Beijing) on scalable dashboards for HPC systems.

- **Road Incidents & Closures Classification in a Connected Vehicular Environment** (CNN, LSTM)

Modeled incident localization as a time-series classification problem using LSTM.

(Full text: <https://shorturl.at/THde8>)

- **Detection of Deep-fakes in Videos using CNN and Transformers** (Deep Learning)

Combined CNN and Transformers to classify manipulated videos in large datasets.

(Full text: <https://shorturl.at/8vACi>)

- **A Comprehensive Review of Convolutional Neural Network-based Image Enhancement Techniques** (CNN, Image Enhancement)

Reviewed image enhancement methods using CNNs for super-resolution tasks.

(Full text: <https://ieeexplore.ieee.org/document/8878706>)

ENGINEERING PROJECTS

- **A Comparative Overview of Dimension Reduction Methods**

Created an explainable notebook comparing PCA, MDS, and t-SNE for Human-AI Interaction course.

(Notebook: <https://observablehq.com/@3e787c1af6c5a437/a-comparative-overview-of-dimension-reduction-methods>)

- **Statistical Analysis of Cricketers**

Developed a novel ODI batsmen ranking system and analyzed performance adjusted for batting era.

(ODI Article: <https://medium.com/@aviseth/who-is-the-greatest-odi-batsman-of-all-time-8add8d896a47>)

- **Autonomous Invoice Processing**

Developed a system for automated processing & database storage of typed/handwritten invoices, winning **SIH 2019**.

(Demo Video: <https://www.youtube.com/watch?v=uq4nnyigLMQ>)

- **Shiny Dashboard for NDVI Dataset and Hampton Roads Analysis**

Visualized vegetation index trends to support policy improvements in Virginia, identified socioeconomic issues faced by African American communities using data visualization..

(Dashboards: <https://aaec-vt.shinyapps.io/dspg21NDVI/> and <https://aaec-vt.shinyapps.io/HamptonRoads/>)

HONORS AND CERTIFICATIONS

- AWS Certified Cloud Practitioner Certification (CLF-C02), AI Practitioner Certification (AIF-C01)

- **Winner** of Government of India's **Smart India Hackathon 2019** at IIT-Roorkee.

- **Winner** of IET's Hack-N-Code 2.0 2018, hosted by HackerEarth.

SKILLS

Programming:	Python, R, C++, GoLang, Java, JavaScript, SQL, Bash
Cloud and DevOps:	AWS (SageMaker, Bedrock, Glue, EC2, Lambda), Docker, Kubernetes
Machine Learning:	XGBoost, Generative AI, Reinforcement Learning, Bayesian Methods
Data Science:	Sk-learn, NumPy, Pandas, Matplotlib, Seaborn, ggplot2, tidyverse
Databases:	MongoDB, NoSQL, PostgreSQL, SAS
Web Development:	Flask, Django, React, Node.js, HTML, CSS, PHP
Tools:	Git, Jira, Confluence, SVN